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compact1, compact2, compact3

java.lang

Class Number

java.lang.Object
    java.lang.Number

All Implemented Interfaces:

Serializable

Direct Known Subclasses:

AtomicInteger, AtomicLong, BigDecimal, BigInteger, Byte, Double, DoubleAccumulator, DoubleAdder, Float, Integer, Long, LongAccumulator, LongAdder, Short

---

```
public abstract class Number
extends Object
implements Serializable
```

The abstract class `Number` is the superclass of platform classes representing numeric values that are convertible to the primitive types `byte`, `double`, `float`, `int`, `long`, and `short`. The specific semantics of the conversion from the numeric value of a particular `Number` implementation to a given primitive type is defined by the `Number` implementation in question. For platform classes, the conversion is often analogous to a narrowing primitive conversion or a widening primitive conversion as defining in *The Java™ Language Specification* for converting between primitive types. Therefore, conversions may lose information about the overall magnitude of a numeric value, may lose precision, and may even return a result of a different sign than the input. See the documentation of a given `Number` implementation for conversion details.

Since:

`JDK1.0`

See Also:

Serialized Form

See *The Java™ Language Specification*:

5.1.2 Widening Primitive Conversions, 5.1.3 Narrowing Primitive Conversions

Constructor Summary

Constructors

Constructor	Description
`Number()`	

Method Summary

Abstract Methods | **Concrete Methods**

Modifier and Type	Method	Description
byte	`byteValue()`	Returns the value of the specified number as a `byte`, which may involve rounding or truncation.
abstract double	`doubleValue()`	Returns the value of the specified number as a `double`, which may involve rounding.
abstract float	`floatValue()`	Returns the value of the specified number as a `float`, which may involve rounding.
abstract int	`intValue()`	Returns the value of the specified number as an `int`, which may involve rounding or truncation.
abstract long	`longValue()`	Returns the value of the specified number as a `long`, which may involve rounding or truncation.
short	`shortValue()`	Returns the value of the specified number as a `short`, which may involve rounding or truncation.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Detail

Number

```
public Number()
```

Method Detail

intValue

```
public abstract int intValue()
```

Returns the value of the specified number as an `int`, which may involve rounding or truncation.

Returns:

`the numeric value represented by this object after conversion to type int.`

longValue

```
public abstract long longValue()
```

Returns the value of the specified number as a `long`, which may involve rounding or truncation.

Returns:

`the numeric value represented by this object after conversion to type long.`

floatValue

```
public abstract float floatValue()
```

Returns the value of the specified number as a `float`, which may involve rounding.

Returns:

the numeric value represented by this object after conversion to type float.

doubleValue

```
public abstract double doubleValue()
```

Returns the value of the specified number as a `double`, which may involve rounding.

Returns:

the numeric value represented by this object after conversion to type double.

byteValue

```
public byte byteValue()
```

Returns the value of the specified number as a `byte`, which may involve rounding or truncation.

This implementation returns the result of `intValue()` cast to a `byte`.

Returns:

the numeric value represented by this object after conversion to type byte.

Since:

JDK1.1

shortValue

```
public short shortValue()
```

Returns the value of the specified number as a `short`, which may involve rounding or truncation.

This implementation returns the result of `intValue()` cast to a `short`.

Returns:

the numeric value represented by this object after conversion to type short.

Since:

JDK1.1

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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